# Rationalising Denominators

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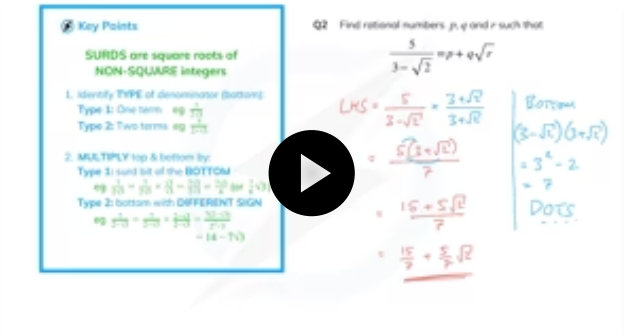


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## Rationalising denominators

### What does rationalising the denominator mean?

- If a fraction has a denominator containing a surd then it has an **irrational** denominator
  - E.g. $\frac{4}{\sqrt{5}}$ or $\sqrt{\frac{2}{3}} = \frac{\sqrt{2}}{\sqrt{3}}$
- The fraction can be rewritten as an equivalent fraction, but with a **rational** denominator
  - E.g. $\frac{4\sqrt{5}}{5}$ or $\frac{\sqrt{6}}{3}$
- The numerator may contain a surd, but the denominator is **rationalised**

### How do I rationalise simple denominators?

- If the **denominator** is a **surd**:
  - Multiply **the top and bottom of the fraction** by the **surd on the denominator**
    - $\frac{a}{\sqrt{b}} = \frac{a}{\sqrt{b}} \times \frac{\sqrt{b}}{\sqrt{b}}$
    - This is equivalent to multiplying by 1, so does not change the value of the fraction

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

simplify if needed

- $\frac{a\sqrt{b}}{b}$

## How do I rationalise harder denominators?

- If the **denominator** is an **expression containing a surd:**
  - For example $\frac{2}{3 + \sqrt{5}}$
  - Multiply **the top and bottom of the fraction** by the **expression on the denominator**, but **with the sign changed**
    - $\frac{2}{3+\sqrt{5}} = \frac{2}{3 + \sqrt{5}} \times \frac{3 - \sqrt{5}}{3 - \sqrt{5}}$
    - This is equivalent to multiplying by 1, so does not change the value of the fraction
  - Multiply the fractions as you would usually (use brackets to help)
    - $\frac{2}{3+\sqrt{5}} = \frac{2(3-\sqrt{5})}{(3+\sqrt{5})(3-\sqrt{5})}$
  - Expand any brackets, and simplify
    - $\frac{2}{3+\sqrt{5}} = \frac{6-2\sqrt{5}}{(3\times 3)+3\sqrt{5}-3\sqrt{5}-(\sqrt{5}\sqrt{5})} = \frac{6-2\sqrt{}}{9-(5)}$
  - You can use the **difference of two squares** to expand the denominator quickly
    - $(a+\sqrt{b})(a-\sqrt{b}) = a^2-(\sqrt{b})^2 = a^2-b$
    - This is what makes the denominator **rational**
  - Simplify
    - $\frac{2}{3+\sqrt{5}} = \frac{6-2\sqrt{5}}{4} = \frac{3-\sqrt{5}}{2}$

### Examiner Tips and Tricks

If your answer still has a surd on the bottom, go back and check your working!

### Worked Example

Write $\frac{4}{\sqrt{6} - 2}$ in the form $p + q\sqrt{r}$ where $p$, $q$ and $r$ are integers and $r$ has no square factors.

**Answer:**

$$\frac{4}{\sqrt{6}-2} = \frac{4}{\sqrt{6}-2} \times \frac{\sqrt{6}+2}{\sqrt{6}+2}$$

Multiply the fractions as you would usually

$$\frac{4}{\sqrt{6}-2} = \frac{4(\sqrt{6}+2)}{(\sqrt{6}-2)(\sqrt{6}+2)}$$

Expand the brackets
The denominator can be expanded using the difference of two squares

$$\frac{4}{\sqrt{6}-2} = \frac{4\sqrt{6}+8}{(\sqrt{6})^2-(2)^2} = \frac{4\sqrt{6}+8}{6-4}$$

Simplify

$$\frac{4}{\sqrt{6}-2} = \frac{4\sqrt{6}+8}{2} = 2\sqrt{6}+4$$

Write in the form given in the question

$$4 + 2\sqrt{6}$$
$$p = 4$$
$$q = 2$$
$$r = 6$$

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